Design review #2/3 – Temperature sensors

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Revision:	A

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1 General information

This project will develop an instrumentation amplifier (INA) with programmable gain (through I2C / SPI interfaces) which achieves high accuracy(< 1% gain error), low noise (< 50nV/rtHz at 1kHz), and high common mode rejection (> 80dB). Design reviews #2 and #3 for this are described in [1] and [2] respectively. Two on-chip temperature sensors are included for accurate determination of the die temperature. One of these is read by the INA itself with the other being used for characterization purposes. This doc deals with the DR2/3 treatment of these temp sensors.

1.1 Project goal

To show the capability of open source tools to develop industry quality ASICs, verified across industry standard conditions and taped out on silicon from a European fab (IHP).

To validate this goal, targeted performance will match that of an INA from a leading industry vendor (AD8223A [3]).

1.2 Specifications

Temperature readings at the output of the INA must be accurate to ±1C.

1.3 Purpose of Design review #2 (DR2) and Design review #3 (DR3)

Purpose of DR2 is to review final schematics before delivery to layout. Therefore, upon approval of DR2, schematics are “frozen” and layout commences.

Purpose of DR3 is to review final layouts before delivery to top level (i.e. chip level) integration. Therefore, upon approval of DR3, layouts are “frozen” and top level integration commences. Specifically, this DR3 refers to the layout freeze of the INA top level (layout freezes for blocks outside of the INA top level are detailed in separate DR3’s).

1.4 Repository details

All material relating to this project can be found in the below publicly available repository:

https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main

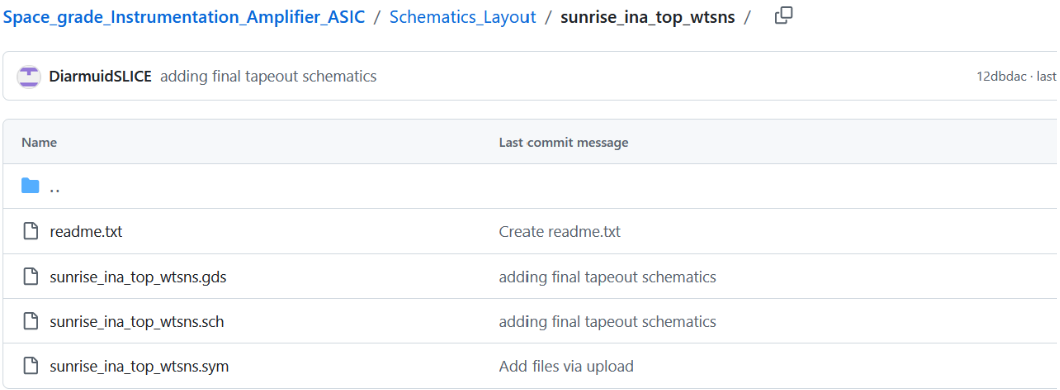
As this document pertains to both DR2 and DR3, all material relating to it is found in the sub directory “Design_Review_3” (folder = TSENSE).

As block schematics relate to all DR’s, these are contained in the separate folder “Schematics_Layout”.

Inside each folder here, lives the following views:

- Schematic (to be frozen at DR2)
- Symbol (to be frozen at DR2)
- Layout (to be frozen at DR3)

Checked in schematic / symbol / layout for the top level where the temp sensors are instantiated is shown in Figure 1 while that for the individual temp sensors is shown in Figure 2.


Space_grade_Instrumentation_Amplifier_ASIC / Schematics_Layout / sunrise_ina_top_wtsns /

DiarmuidSLICE adding final tapeout schematics 12dbdac · last

Name	Last commit message
..	
readme.txt	Create readme.txt
sunrise_ina_top_wtsns.gds	adding final tapeout schematics
sunrise_ina_top_wtsns.sch	adding final tapeout schematics
sunrise_ina_top_wtsns.sym	Add files via upload

Figure 1 Checked in schematics for top level instantiation of temp sensors

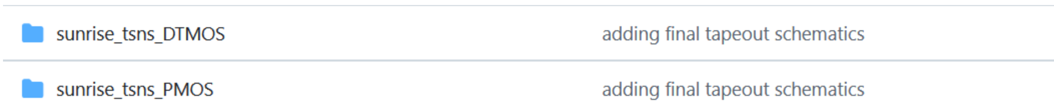

sunrise_tsns_DTMOS	adding final tapeout schematics
sunrise_tsns_PMOS	adding final tapeout schematics

Figure 2 Checked in schematics for temp sensors

1.5 Open source tools

The open source tools used for DR2 are summarized in Table 1, with links to install each tool provided as references.

Tool	Description
ngspice	Spice simulator [4]
xschem	Schematic / waveform viewer [5]
openVAF	verilogA compiler [6]
Octave	Post processing [7]
klayout	Layout viewer / editor [8]
magic	Pcell generator / PEX generation [9]
netgen	LVS [10]

Table 1 Open source tools used for DR2/3

1.6 Open source PDK

The open source process development kit (PDK) used for the design is from IHP, process option – SG13CMOS5L [11]. This is a derivative of their more general option – SG13G2 [12] as it excludes the following:

- Top metal layer 2
- Linear caps (MiM or MoM)
- Deep Nwell (DNW)
- HBT devices

The first 3 items impacted the design but not the 4th since the design is not RF. SG13CMOS5L option was chosen primarily because its tapeout date of 13/07/26 suited the project timeline. In addition, it is lower cost (max = €1.5k/mm2 as opposed to €2.8k/mm2 for SG13G2) and runs through the fab quicker (5 months instead of 6) to achieve a fab out date in 16/12/26 [13].

2 Temp sensor theory

Accurate temperature sensors are always based on some physical property which varies linearly with temperature. The most common property used is that of thermal voltage. As shown below, this varies linearly with temperature at a rate of 86uV/C.

$$v_t = \frac{kT}{q} => \frac{d(v_t)}{dT} = \frac{k}{q} = 86\mu V/°\mathrm{C} \ldots (1)$$

This property is most commonly extracted in circuit design by measuring the difference in base-emitter voltages of 2 BJTs (biased by a current ratio = M) to give:

$$\Delta v_{be} = n.v_t.\ln(M) = n.\frac{kT}{q}.\ln(M) => \frac{d(\Delta v_{be})}{dT} = n.\ln(M).\frac{k}{q} = 198\mu V/°\mathrm{C}\ (n = 1, m = 10) \ldots (2)$$

As shown above, assuming $\Delta n = 0$ across process, Δv_{be} varies linearly with temperature at a rate of ~ 200uV/C. In practice $\Delta n \neq 0$ which typically limits the accuracy of such temperature sensors to 1-2C, in the absence of additional techniques (of which there are many[14]).

Instead of using BJTs, this ASIC uses MOSFETs biased in weak inversion to measure temperature using Δv_{gs} as shown below:

$$\Delta v_{gs} = n.v_t.\ln(M) = n.\frac{kT}{q}.\ln(M) => \frac{d(\Delta v_{gs})}{dT} = n.\ln(M).\frac{k}{q} = 198\mu V/°\mathrm{C}\ (n = 1, m = 10) \ldots (3)$$

As shown above, assuming $\Delta n = 0$, Δv_{gs} varies linearly with temperature at a rate of ~ 200uV/C. The reason this ASIC uses Δv_{gs} instead of Δv_{be} is Δv_{gs} remains linear to much lower temperatures (below -200C), since BJTs start to suffer from freeze out effect below -100C[15]. As this ASIC is intended for space applications, it is important to have the capability to measure temperature down to -200C.

Measuring a resolution of ≈ 200μV/C is too small to measure in the presence of circuit non-idealities, Δv_{gs} is sensed by the INA to gain it up by 21x. The INA thus acts as the analog front end (AFE) to the temp sensor, which outputs a voltage that varies with temperature at a rate of 4.2mV/C. This is a much more practical magnitude to measure in the presence of circuit non-idealities and can be achieved with a much less involved ADC.

3 Architecture details

Temp sensors are shown in Figure 3 where they can be seen to comprise of 2 diode connected PMOS devices with same W/L dimensions. Currents into them differ by a factor of 10 (10uA/1uA) to generate M factor = 10. This makes the architecture a current ratioed one, in contrast to a device ratioed one whereby the same current is fed into 2 devices whose area differs by a factor of 10.

Note in Figure 3 the bulk of the PMOS devices is connected to supply (vdda_hv). For research purposes, a 2nd temp sensor architecture has been implemented where the bulks of the PMOS devices are connected to their drains. This is shown in Figure 4 and is referred to as a “Dynamic threshold MOS” or DTMOS. The reason it has been included in the ASIC is it has been reported [16] to exhibit lower Δn across process than a standard PMOS. Since Δn across process is not modelled by IHP (or most foundries for that matter), its spread can only be examined in silicon, which is why the answer to the optimum temp sensor cannot be determined from simulation alone.

Instantiation of the temp sensors in the top level is then shown in Figure 5. From here you can see the PMOS temp sensors are connected directly to the INA input mux, while the DTMOS ones are not. This is because their common mode is too low to be read by the INA. As a result, they exist only as standalone devices on the die, purely for post silicon characterisation purposes.

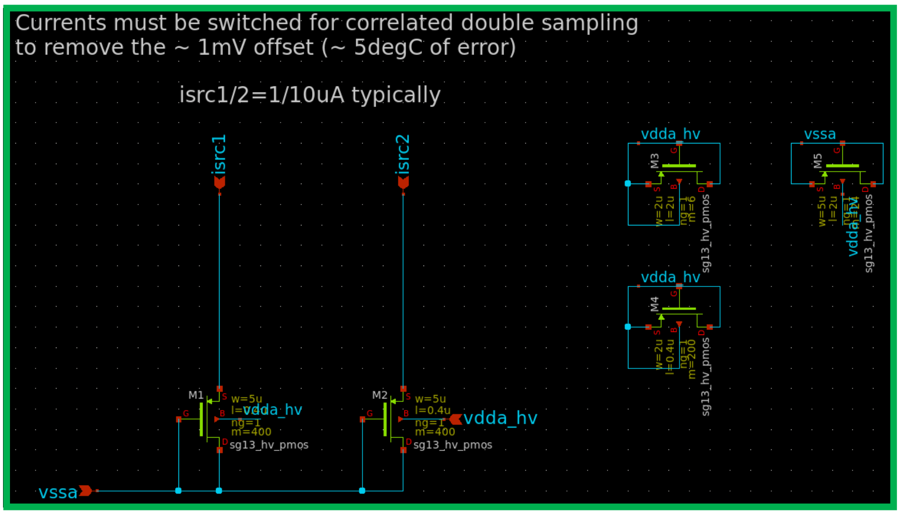


Figure 3 PMOS temp sensors

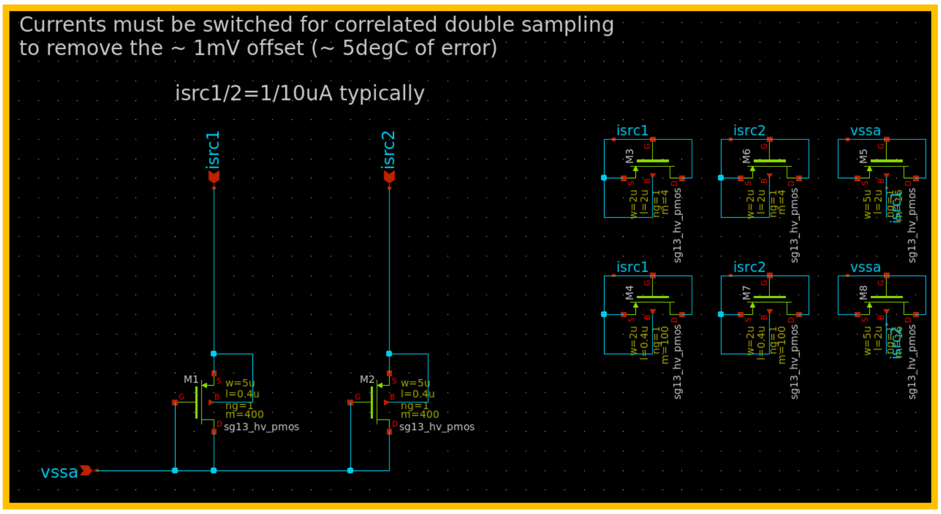


Figure 4 DTMOS temp sensors

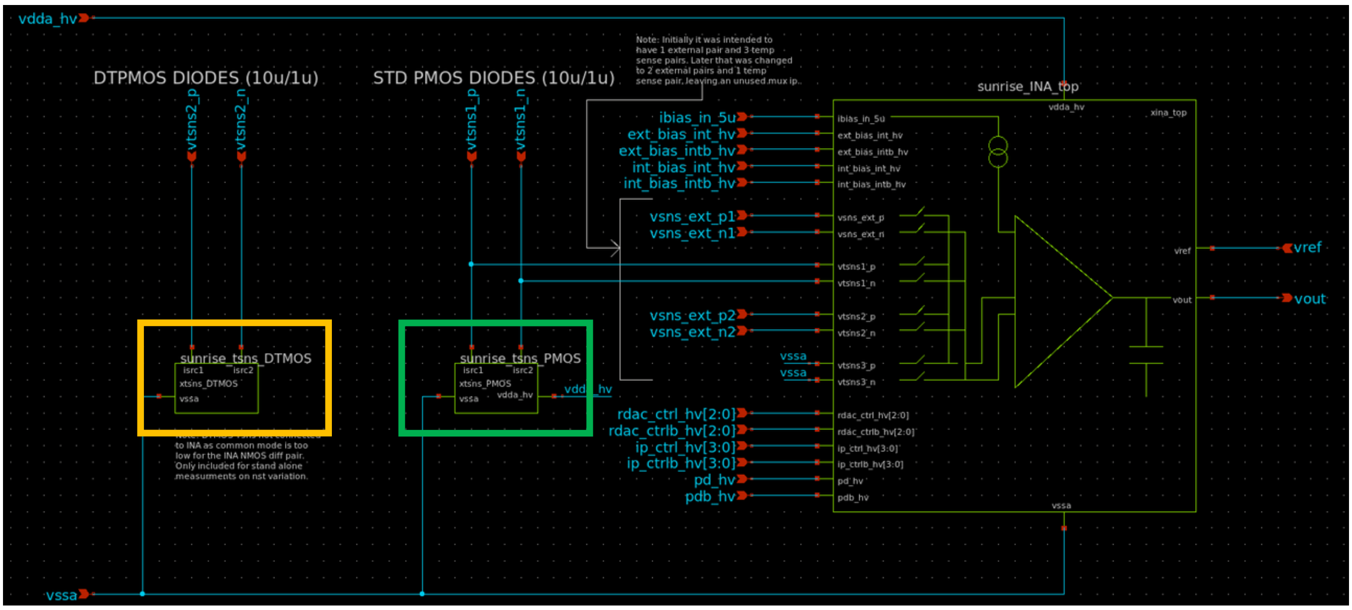


Figure 5 Top level temp sensor instantiation

4 Temp sensor characterisation

In order to be used, $T(\Delta v_{gs})$ of the temp sensor must be characterised, which in practice translates to generating the 2 coefficients describing its linear transfer function. Characterisation from -40 → 85C is detailed in this section.

4.1 PMOS diodes

Characterisation set up is shown in Figure 6, where 2 currents (10/1uA) are sourced into the drains of the diodes.

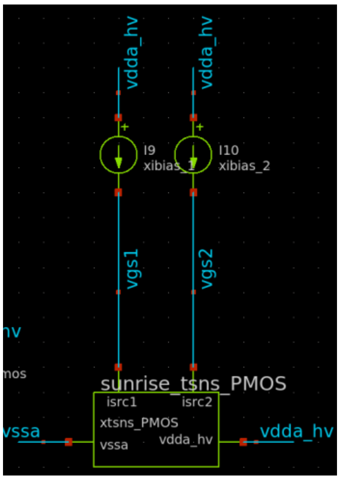


Figure 6 PMOS diode characterisation setup

Resulting plots of vgs1,2 and dvgs across temperature are shown in Figure 7and Figure 8 respectively.

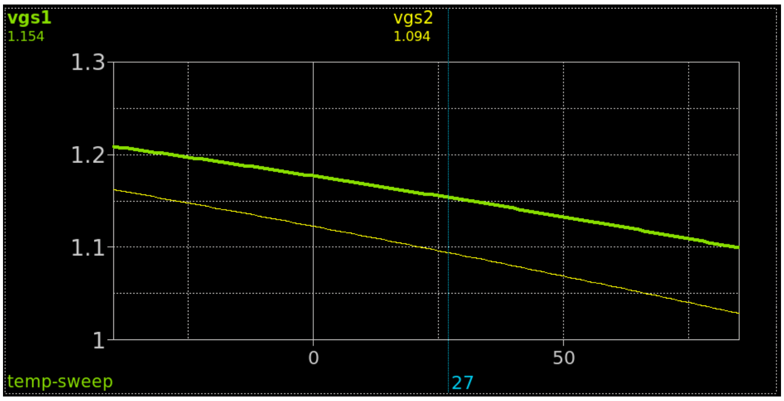


Figure 7 vgs1,2 across T (($\Delta v_{gs_{1,2}}$(T))

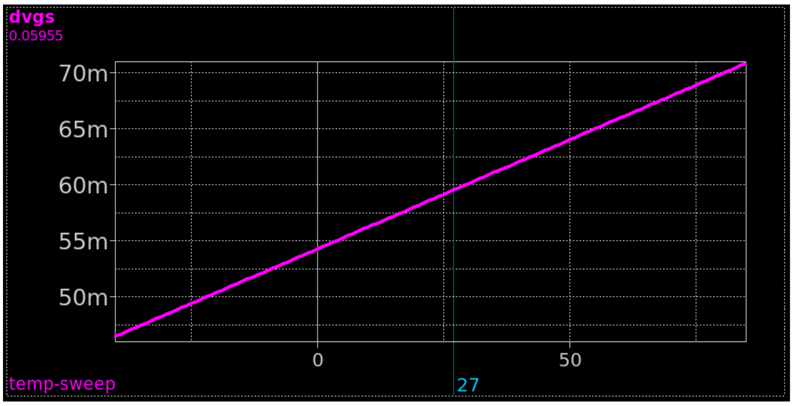


Figure 8 dVgs across T ((Δv_{gs}(T))

To characterise $T(\Delta v_{gs})$ we place Δv_{gs} on the x-axis and T on the y-axis. This is done in octave to generate the "char" straight line shown in Figure 9.

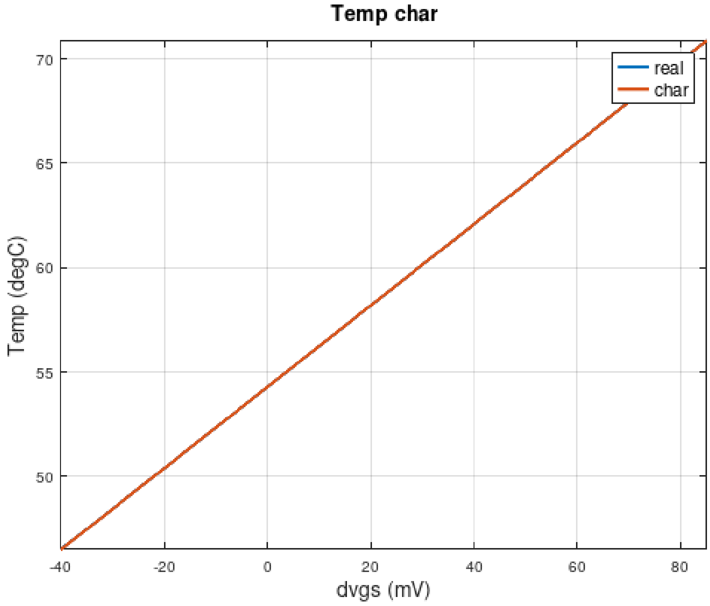


Figure 9 T(∆vgs) and accuracy characterisation

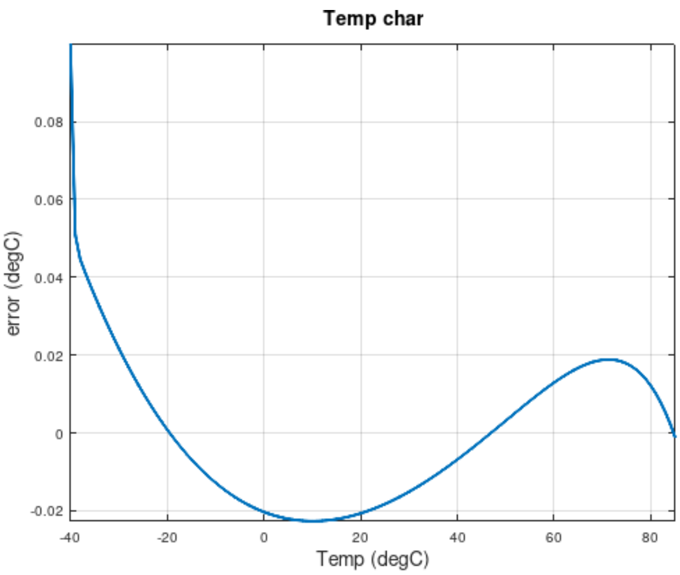
Resulting linear coefficients derived from a standard polynomial fit of this are shown below:

```
m_char: 5131.4
os_char: -278.597
```

Note that taking the inverse of the slope of this gives $1/5131.4 = 194.88\mu V/℃$, which agrees well with the ~ 200uV/C number predicted in section 2.

Using these linear coefficients, we can determine the accuracy of the standalone PMOS diodes by generating a 2nd straight line that calculates T from Δv_{gs}, as done for the "real" straight line shown in Figure 9.

The difference between the 2 straight lines of Figure 9 gives the accuracy of the standalone diodes. This is plotted in Figure 10 which shows a temp sensor accuracy of < 0.1C.




```
max error: 0.0998821
Temp for max error: -40
```

Figure 10 Accuracy of PMOS standalone diode temp sensors

4.2 DTMOS diodes

An identical characterisation to that performed in section 4.1, is performed on the DTMOS diodes.

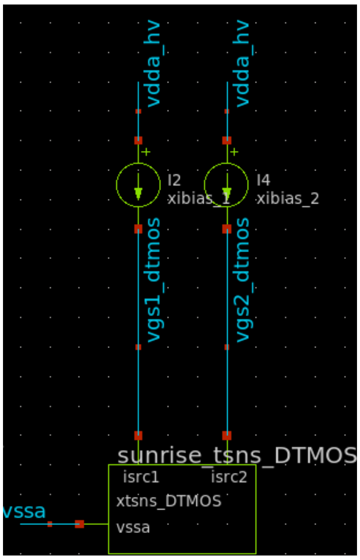


Figure 11 DTMOS diode characterisation setup

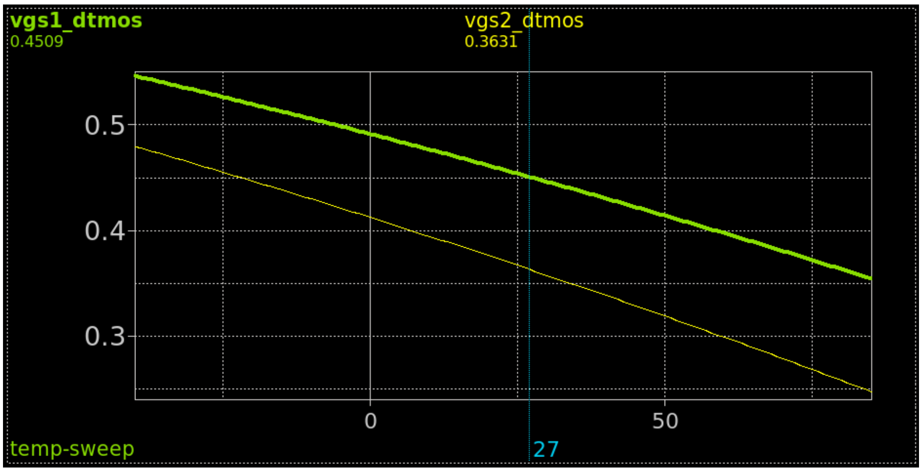


Figure 12 vgs1,2 across T ((Δv_(gs_1,2)(T))

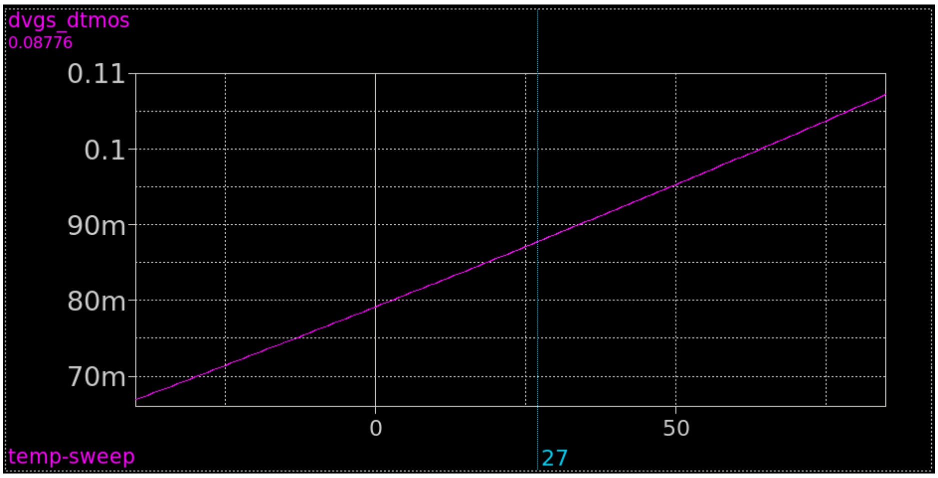


Figure 13 dVgs across T ((Δvgs(T))

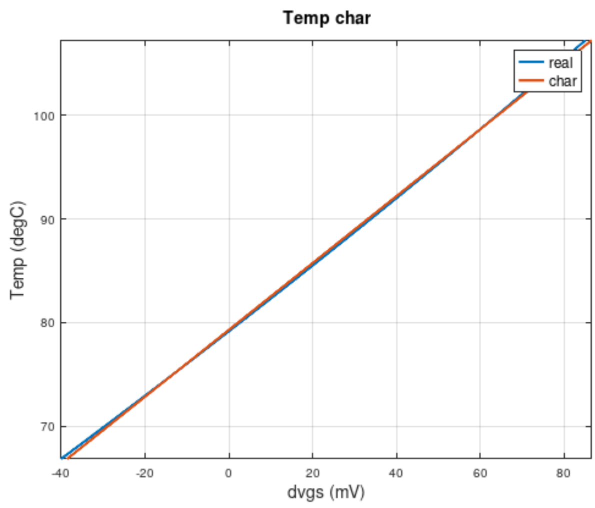


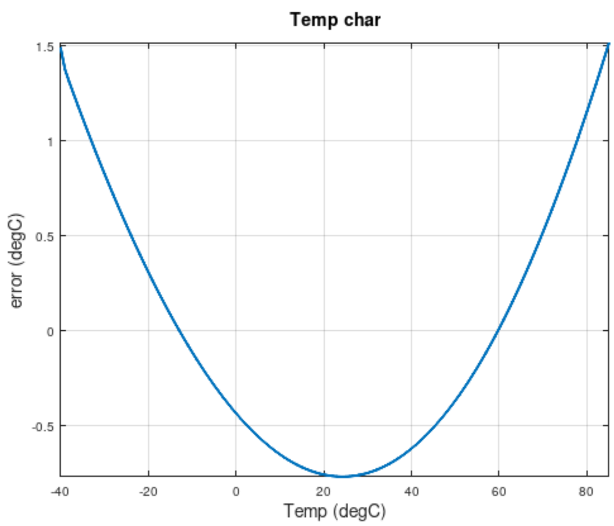
Figure 14 T(∆vgs) and accuracy characterization

Characterised linear coefficients from the "char" plot of Figure 14, as shown below:

```
m_char: 3102.12
os_char: -246.01
```

Interestingly, by taking the inverse of the slope we get $1/3102.12 = 322.36\mu V/°C$, which is much larger than the ~ 200uV/C number predicted for a PMOS sensor in section 2. This indicates either some other effect is occurring or even a modelling issue. Post silicon results will be required to investigate further.

Comparing Figure 14 with Figure 9 one can see the "real" and "char" lines deviate more at the temperature extremes. This implies reduced accuracy at -40 and 85C, as confirmed in Figure 15, which shows a relatively poor temp sensor accuracy of ~ 1.5C.




```
max error: 1.51549
Temp for max error: 85
```

Figure 15 Accuracy of DTMOS standalone diode temp sensors

This represents 15x worse accuracy than the PMOS diodes, which from simulation results alone, look by far to be the optimum temp sensor architecture.

Post silicon results will be eagerly awaited to validate further.

5 Top level performance

As explained in section 2, the INA (with gain = 21) acts as the AFE to the PMOS temp sensor. With the INA configured like so, its resulting output across temperature is shown in Figure 16.

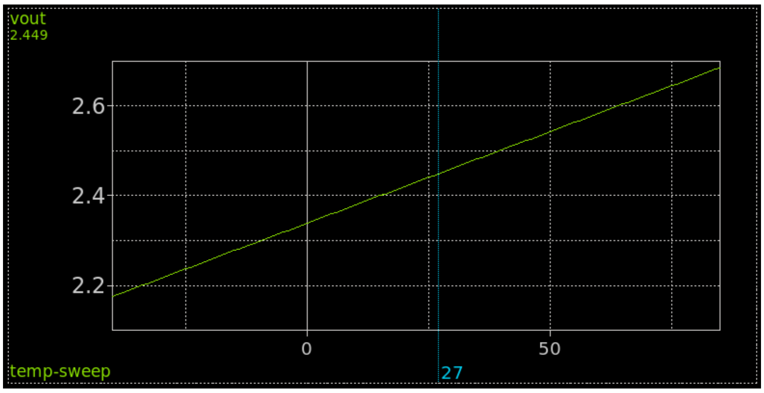


Figure 16 vout across T (V(T))

As a quick sanity check, the cursor in Figure 16 is placed at 27C to measure vout = 2.449. When we remove the output common mode of 1.2 and compare this with the cursor measurement in Figure 8 (59.55mV), we obtain a ratio of $1.249/59.55m$, implying a gain of ~ 21 (not exactly 21 due to gain error, detailed later).

Due to the INA's gain error, characterisation of vout(T) must be derived from $T(\Delta v_{gs})$ in section 4.1. This is achieved by dividing the slope of $T(\Delta v_{gs})$ by the INA gain of 21, to give the below slope (m_char_opr):

```
m_char: 5131.61
os_char: -278.609
m_char_opr: 244.362
```

Using "m_char_opr", the accuracy of the temp sense + INA AFE, can be obtained from Figure 17.

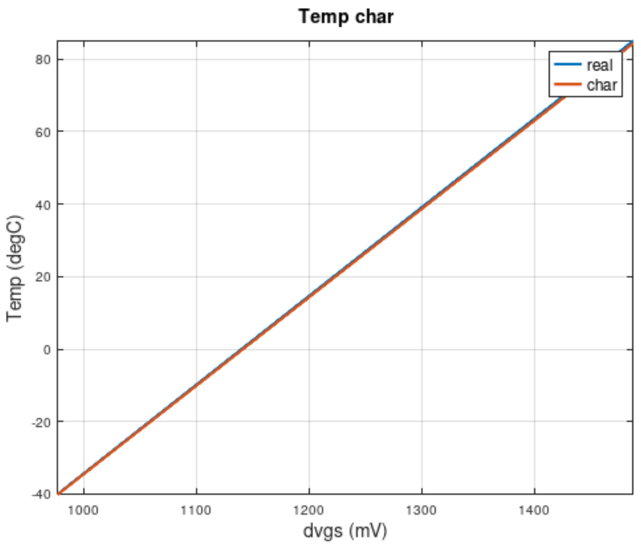


Figure 17 T(Δvout) and accuracy characterization

Again by taking the inverse of the slope of this gives $1/244.36 = 4.1mV/°C$, which agrees well with the ~ 4.2mV/C number predicted in section 2.

Taking the difference between “real” and “char” plots in Figure 17 gives the total error shown in Figure 18.

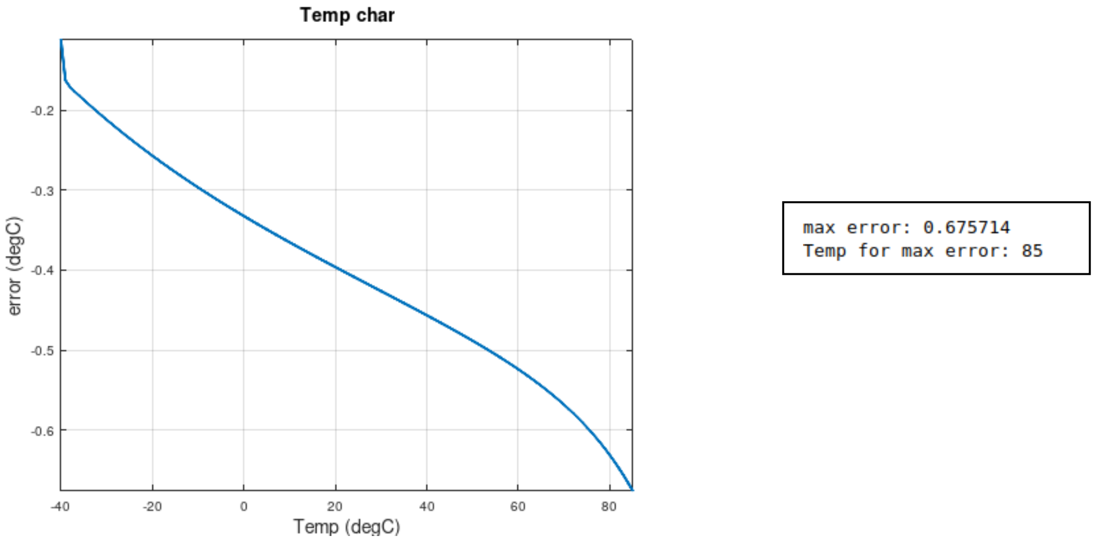


Figure 18 Accuracy of PMOS temp sensors + AFE

Immediately obvious is the error has increased to ~ 0.7C. This is entirely due to the gain error (GE) of the INA as verified with the below calculations:

- Max error occurs at 85C. Using (3), Δv_{gs} at this corner is:

$$\Delta v_{gs} = 1.\ln(10).\left(\left(1.38xe^{-23}.(273+85)\right)/1.6xe^{-19}\right) = 71.1mV$$

- For an INA gain of 21, the resulting Δv_{out} (ignoring GE) is:

$$\Delta v_{outINA_{id}} = 21.\Delta v_{gs} = 1.494V$$

- GE (shown corner) = 0.17%. This will produce the following output error:

$$\Delta v_{out_e} = 1.494x(0.17/100) = 2.54mV$$

- Dividing the 4.1mV/C number into this thus gives the output error in C as:

$$\text{At } d(v_{outINA})/dT = 4.1mV/℃ \Rightarrow T_e = (2.54/4.1) = 0.62℃$$

In addition to root causing the error in Figure 18, the above analysis also confirms our understanding of the system and that everything is behaving as expected.

One could argue over the use of the INA since at a first glance, all it has done was increase the error from 0.1 → 0.7C. To answer this question, one must remember that the output must be measured by an ADC to digitize the result and provide a final result. An ADC required to resolve 200uV/C will be much more complex than one required to resolve 4.1mV/C. Therefore, by sacrificing an additional 0.6C of error we dramatically simplify the ADC requirements and hence, design. This is an important step since a follow on project to this will be to integrate the ADC and INA on the same die.

Finally, the 0.6C number is perfectly acceptable since the accuracy spec set for this temperature sensor is ±1C.

6 Layout analysis

6.1 PMOS diodes

Layout view of the PMOS diodes is shown in Figure 19, showing it to occupy a footprint of ~ 100x100um.

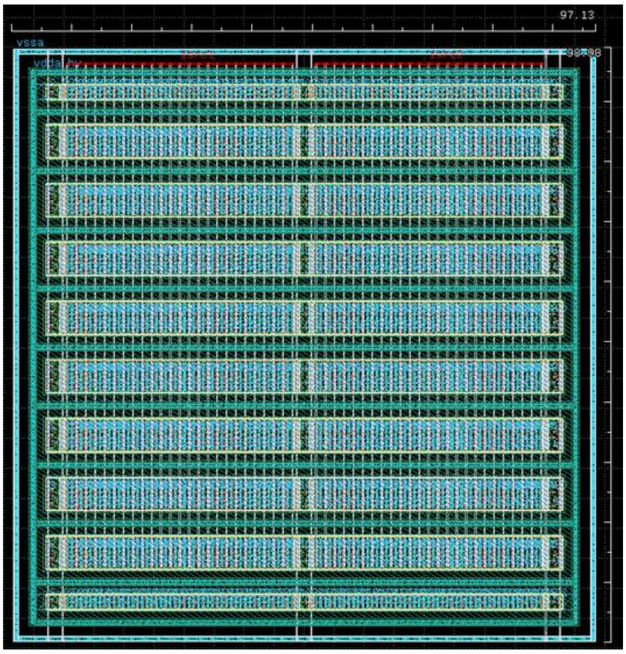


Figure 19 Layout view of PMOS diodes

Pre and post layout views for $\Delta v_{gs}(T)$ and $v_{gs}(T)$ are shown in Figure 20.

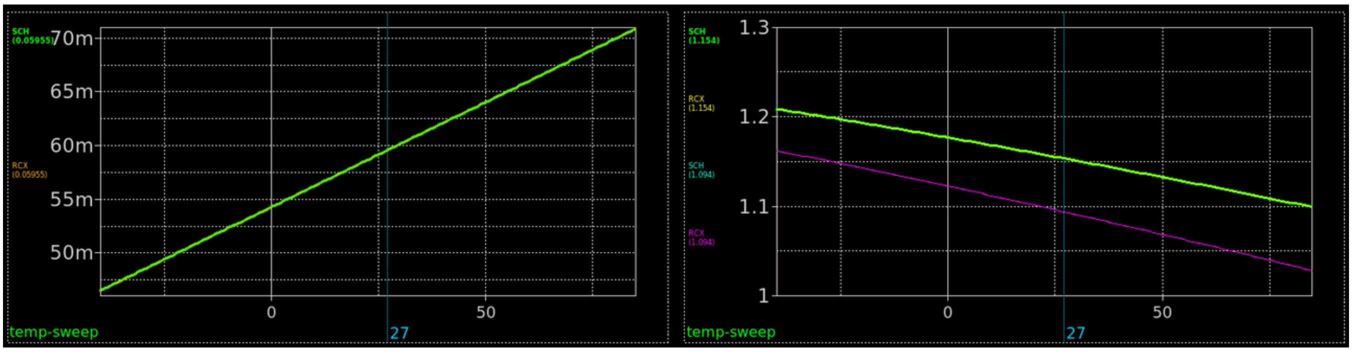


Figure 20 Pre (SCH) and post (RCX) views for PMOS diode characterization

As can be seen, pre and post layout results match almost exactly. As a result, temp sensor characterization and accuracy shown in Figure 21, matches that of Figure 9 and Figure 10.

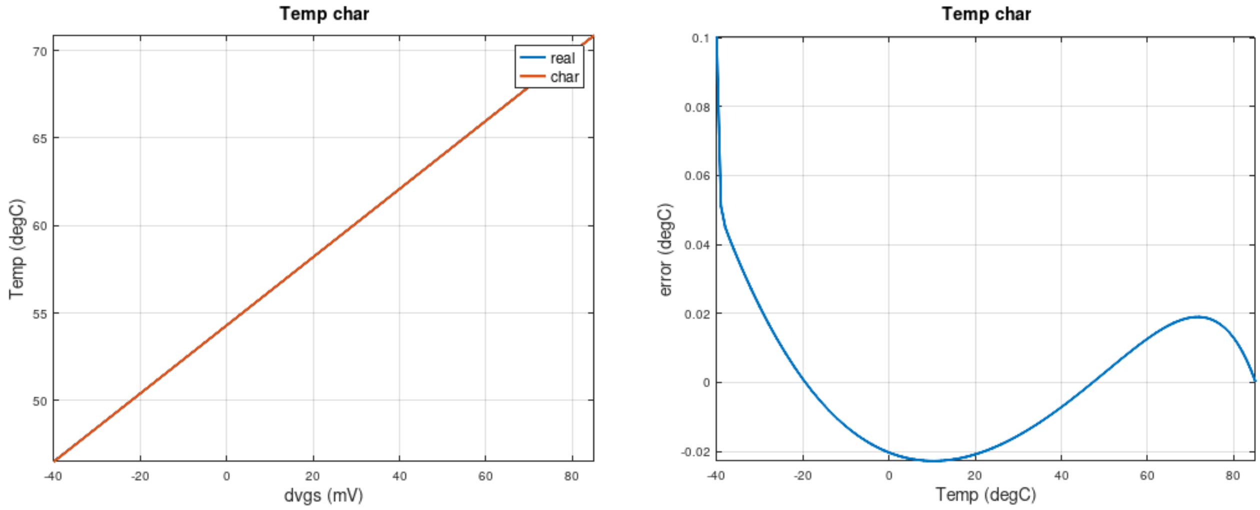


Figure 21 Characterisation and accuracy of PMOS diode temp sensors

6.2 DTMOS diodes

Layout view of the DTMOS diodes is shown in Figure 19, showing it to occupy a footprint of ~ 100x100um.

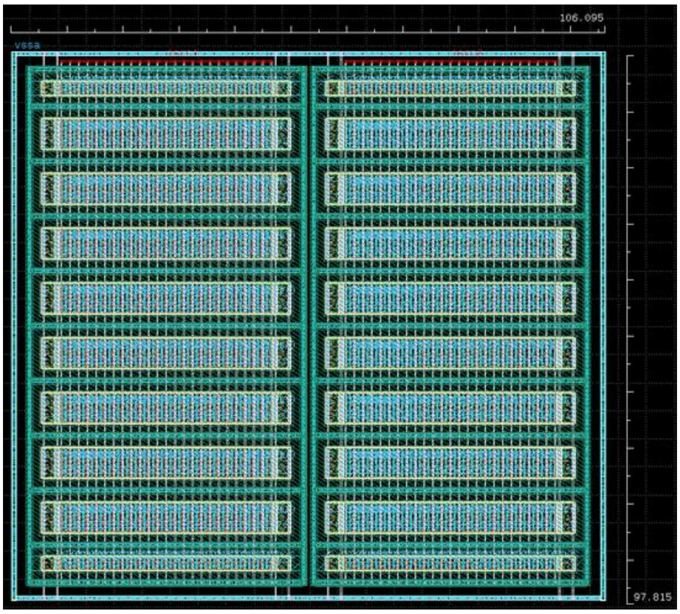


Figure 22 Layout view of DTMOS diodes

Pre and post layout views for $\Delta v_{gs}(T)$ and $v_{gs}(T)$ are shown in Figure 20.

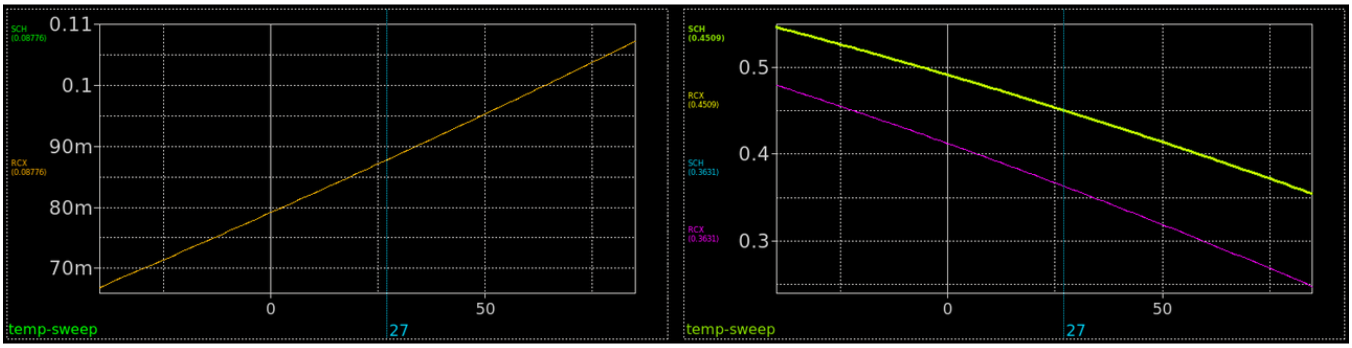


Figure 23 Pre (SCH) and post (RCX) views for DTMOS diode characterization

As can be seen, pre and post layout results match almost exactly. As a result, temp sensor characterization and accuracy shown in Figure 24, matches that of Figure 14 and Figure 15.

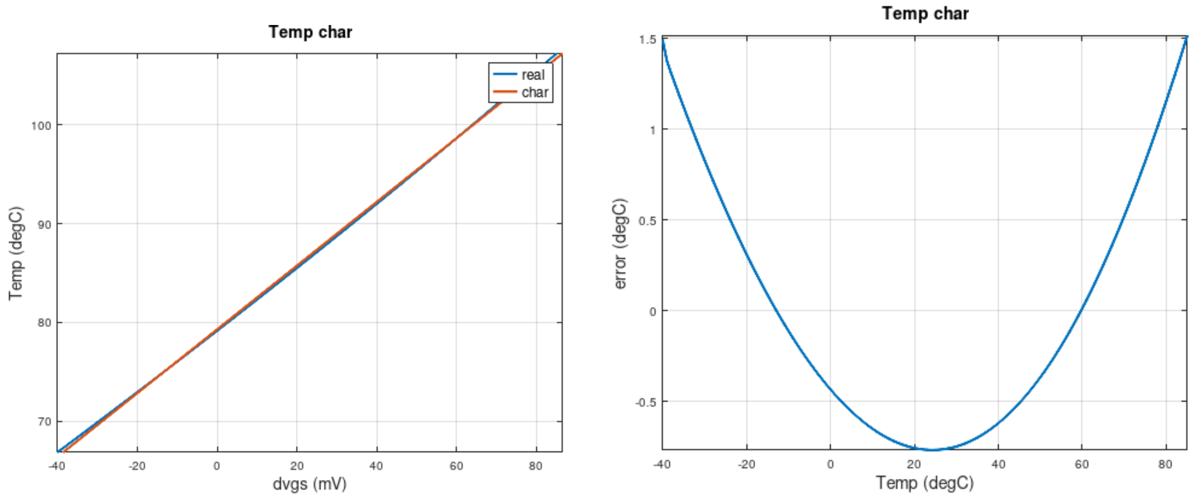


Figure 24 Characterisation and accuracy of DTMOS diode temp sensors

6.3 Top level integration

Temp sensor diode integration into the ASIC top level is shown in Figure 25 → Figure 27.

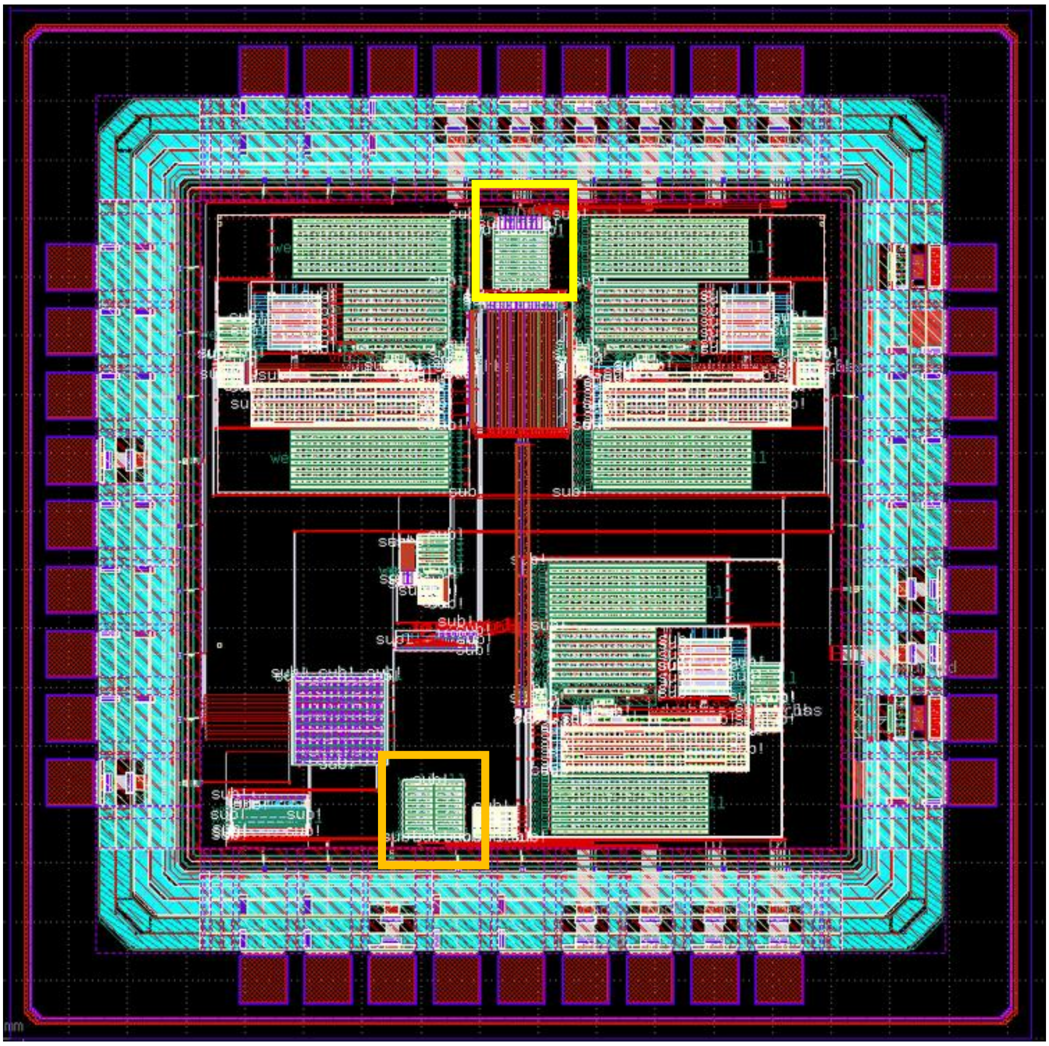


Figure 25 Top level integration of temp sensors

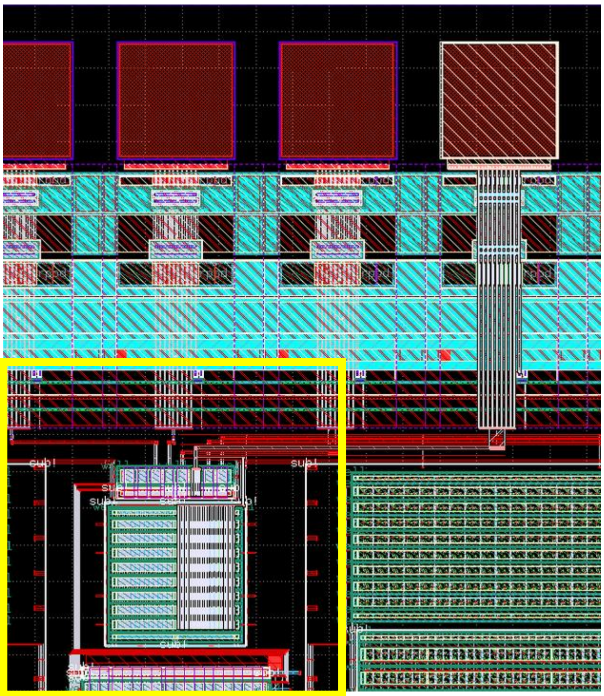


Figure 26 Close up view of PMOS diode integration

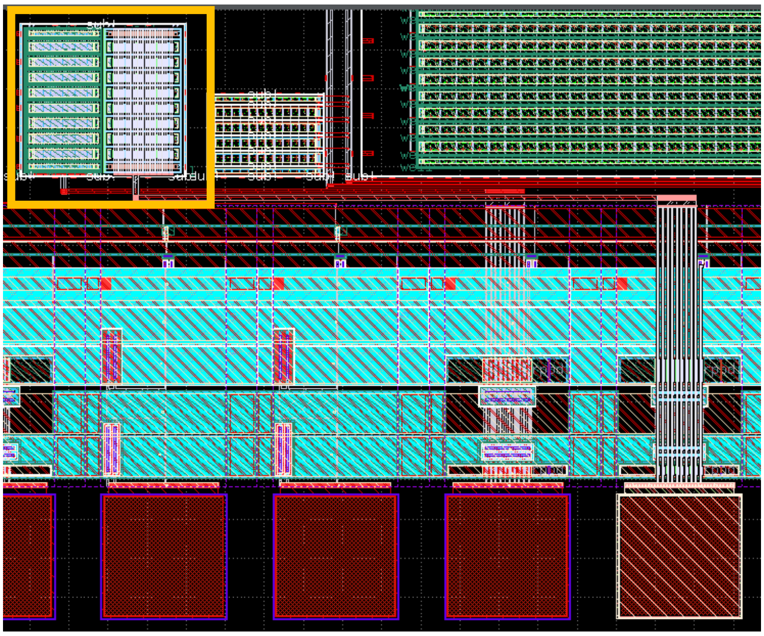


Figure 27 Close up view of DTMOS diode integration

The INA output pre (SCH) and post (RCX) layout across temperature is shown in Figure 28.

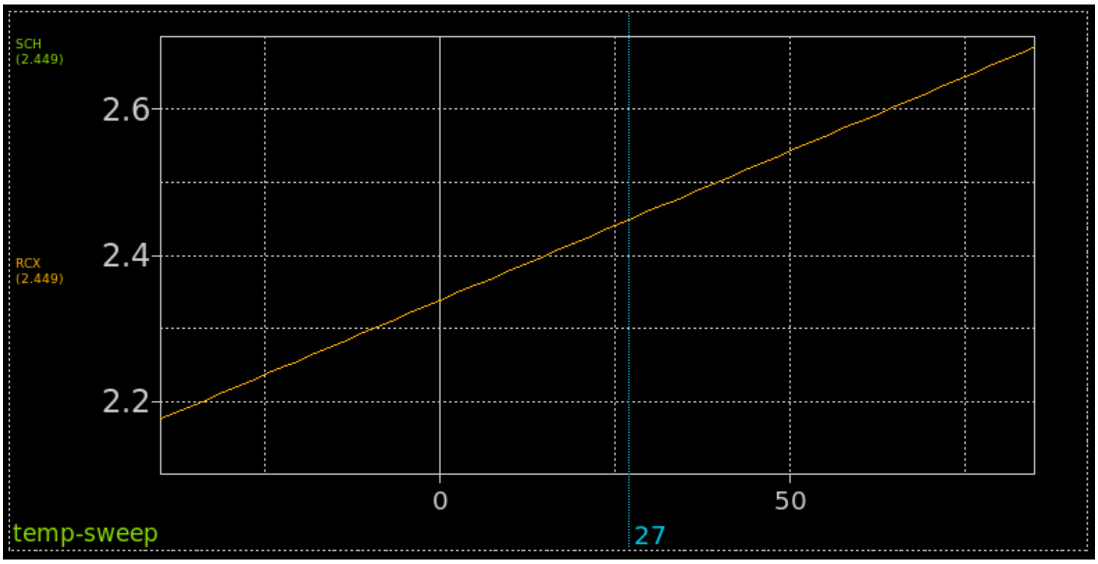


Figure 28 INA output pre and post layout

As can be seen, pre and post layout results match almost exactly.

Using “m_char_opr” from section 5, the post layout accuracy of the temp sense + INA AFE, can be obtained from Figure 29.

Taking the difference between “real” and “char” plots in Figure 29 gives the total error shown in Figure 30. Again this shows the 0.6C error due to gain error, verifying pre and post layout accuracy to match.

One will notice 3 glitches in the waveform of Figure 30. Such sudden glitches are not real and caused due to discontinuities in the simulation. Although it is not ideal to see them, further debug was out of the scope of this work. For future reference, fixing this would require optimising simulation parameters for this particular netlist which, being such a large netlist that takes hours to sim, make this a very time-consuming task.

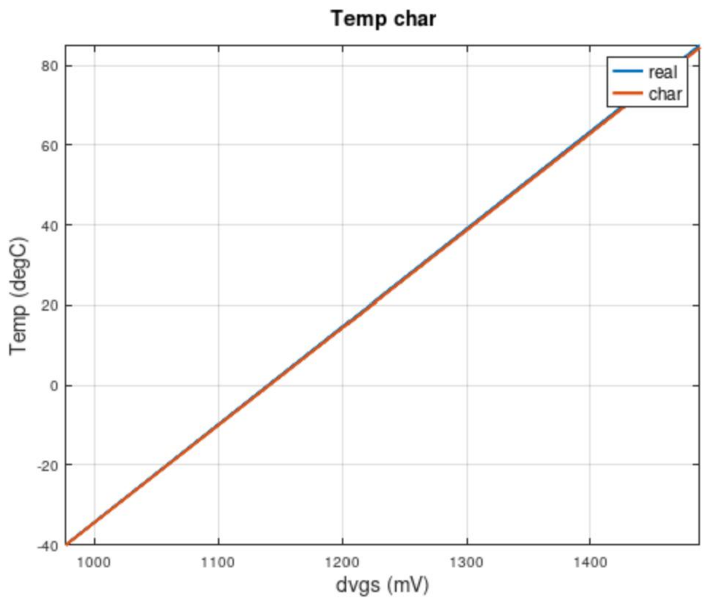
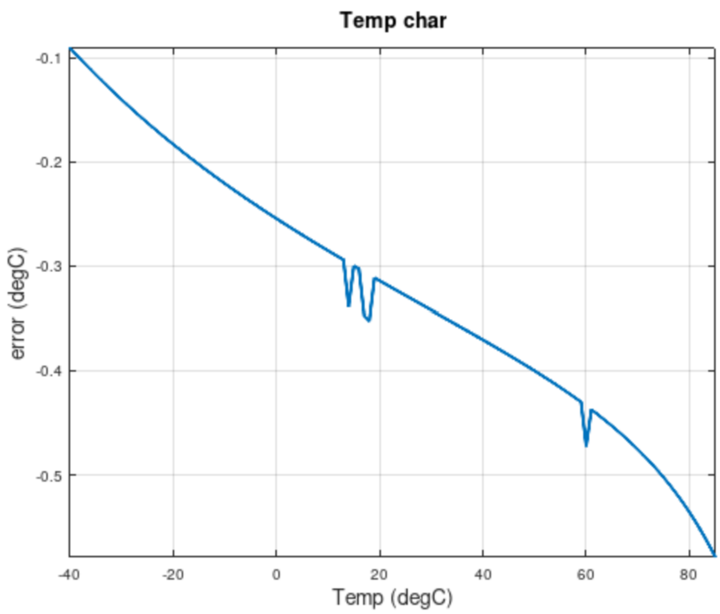


Figure 29 T(∆vout) and accuracy characterization




```
max error: 0.578773
Temp for max error: 85
```

Figure 30 Accuracy of PMOS temp sensors + AFE

7 Summary

An MOS based temperature sensor has been integrated onto the ASIC which uses the INA as the AFE to provide temperature readings to ±1C accuracy, as verified with post layout simulations.

This temp sensing capability can be maintained down to -200C making it highly relevant to the space applications intended for this ASIC.

A 2nd temp sensor has also been included for research purposes, which is not senses by the INA.

8 References

[1] https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main/Design_Review_2/INA

[2] https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main/Design_Review_3/INA

[3] https://www.analog.com/media/en/technical-documentation/data-sheets/AD8223.pdf

[4] https://ngspice.sourceforge.io/download.html

[5] https://xschem.sourceforge.io/stefan/xschem_man/install_xschem.html

[6] https://openvaf.semimod.de/

[7] https://octave.org/download

[8] https://www.klayout.de/build.html

[9] https://opencircuitdesign.com/magic/download.html

[10] https://opencircuitdesign.com/netgen/

[11] https://github.com/IHP-GmbH/ihp-sg13cmos5l

[12] https://github.com/IHP-GmbH/IHP-Open-PDK

[13]https://www.ihp-microelectronics.com/services/research-and-prototyping-service/mpw-prototyping-service/schedule-price-list

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9 Revision History

Rev.	Date	Author	Changes, Remarks
A	16.08.2026	Diarmuid Collins	Created

Table 2: Revision history